

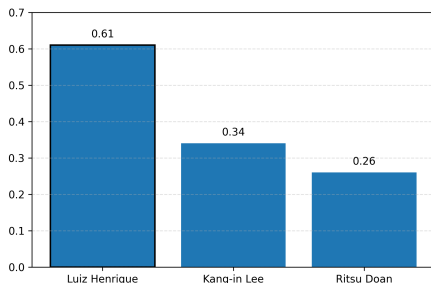
Replacing Riyad Mahrez (Al-Ahli) Multidimensional Statistical Analysis and Recruitment Shortlist

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Executive Summary



Objective

Anticipate the **Mahrez succession plan** (end of peak cycle, 34).

Data-driven comparison of 40 left-footed inverted wingers.

10 normalized metrics linked to Mahrez's role pattern : xA, key passes, progressive passes/carries, xG, touches in box.

Similarity measured using **cosine distance**.

Clear outcome : Luiz Henrique is the closest (**0.61**).

Feasible acquisition : market value $\leq$ €25M.

Recommendation

Prioritized Target : Luiz Henrique

Structurally closest match to Mahrez with immediate tactical transferability and long-term upside.

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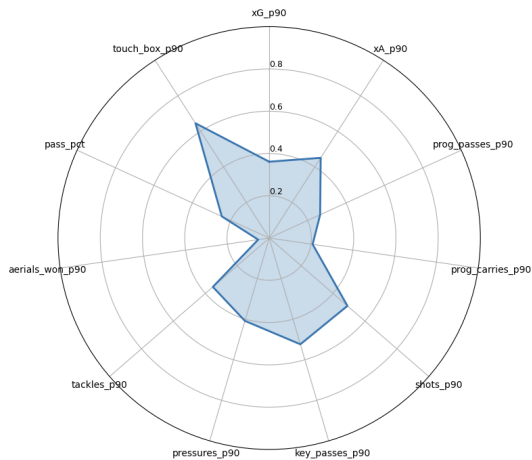
Riyad Mahrez approaching the end of his elite-level cycle (34) : need to anticipate succession planning at **Al-Ahli**.

Objectif : identify a **a statistically comparable profile**, tactically transferable and financially feasible.

Target player profile with similar attributes :

- Impactful in the right half-space (key passes, xA).
- Strong unbalancing ability (progressive passes/runs).
- Player with high involvement inside the penalty area.

Mahrez — Key Profile Summary



Mahrez radar — core metrics (2022/23).

Role archetype

Elite **chance creator** in the right half-space.

High **ball progression** (progressive passes & carries).

Above-average **presence inside the box**.

Key metrics monitored

Creativity : xA_p90, key_passes_p90.

Threat : xG_p90, touches_box_p90.

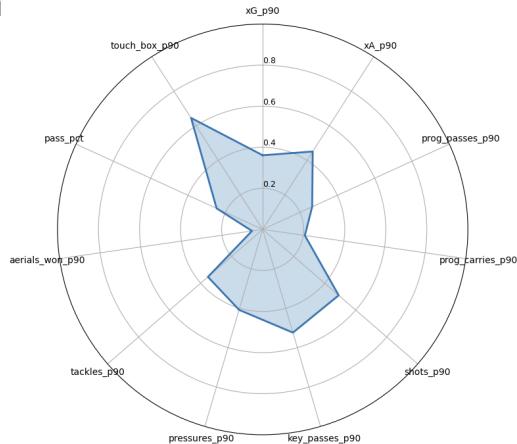
Progression : progressive_passes_p90, progressive_carries_p90.

Statistical Reference Profile — Mahrez (2022/23)

Creator in the right *half-space*, a **consistent playmaker** (xA, key passes).

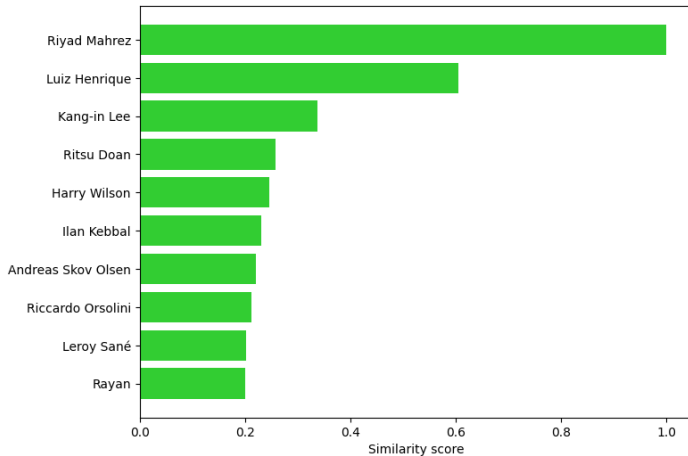
Strong ability to **progress** the ball (progressive passes/carries).

Above-average touches inside the box.



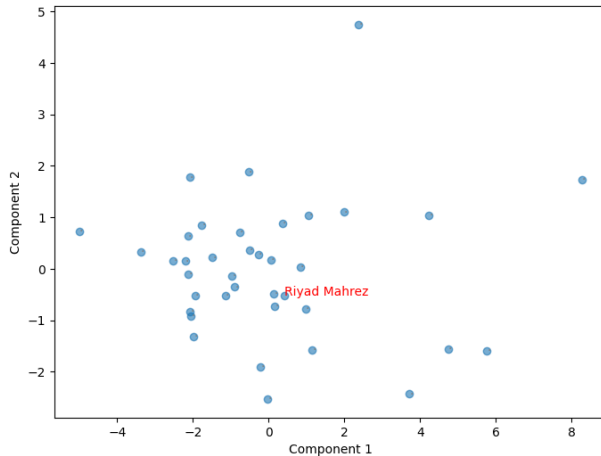
Mahrez radar (baseline).

Top Similarity to Mahrez Profile (Cosine Distance)



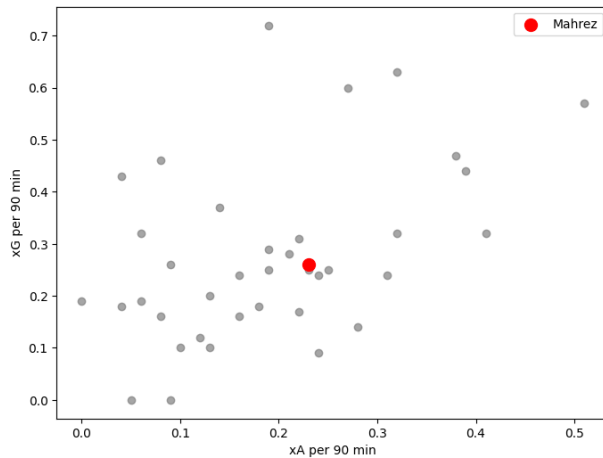
Ordered cosine similarity scores.

PCA Projection — Profile Clusters



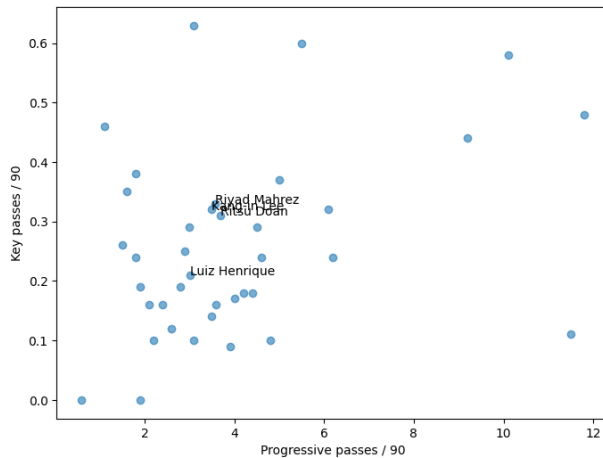
2D PCA projection — structural proximity between profiles.

Creativity vs Threat



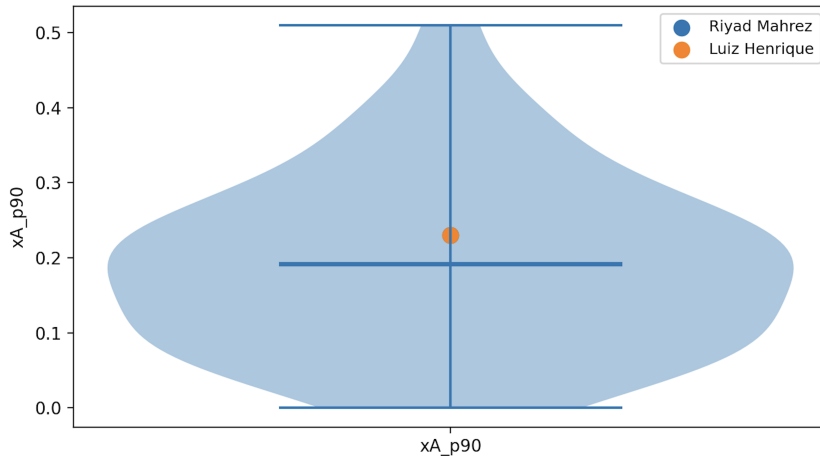
xG/xA positioning — Mahrez vs sample.

Creation vs Progression



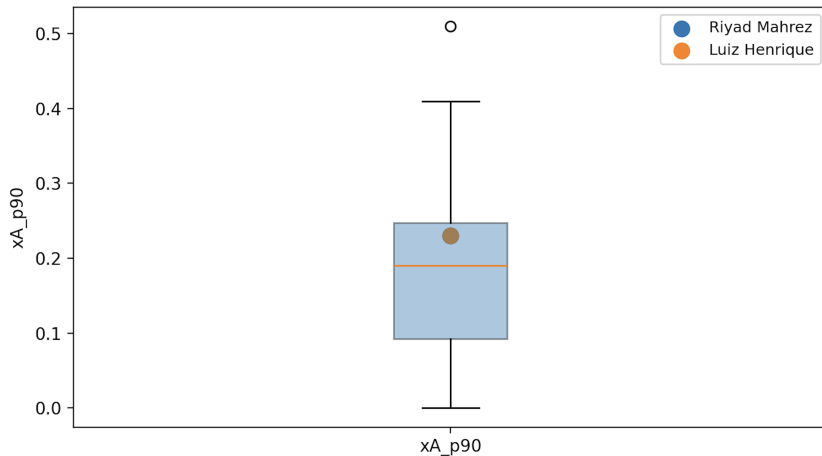
Key passes / progressive passes.

Creation Distribution (xA_p90) — Violin Plot



Creation distribution (xA_p90) — Mahrez vs Henrique.

Creation Variability (xA_p90) — Boxplot



Mahrez and Henrique within the global distribution.

Podium — Most Similar Profiles



Riyad Mahrez

Baseline (2022/23)



Luiz Henrique

Similarity 0.61

1



Kang-in Lee

Similarity 0.34

2

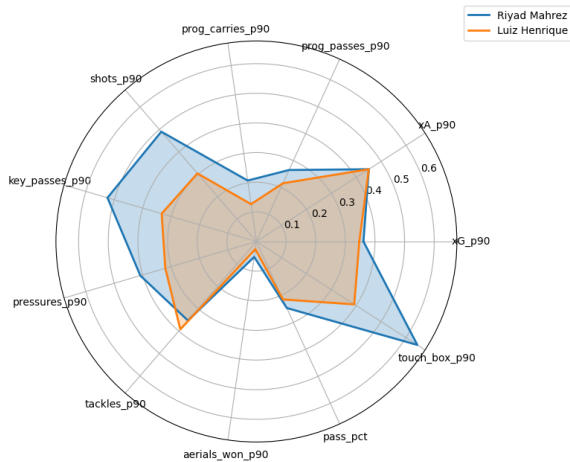


Ritsu Doan

Similarity 0.26

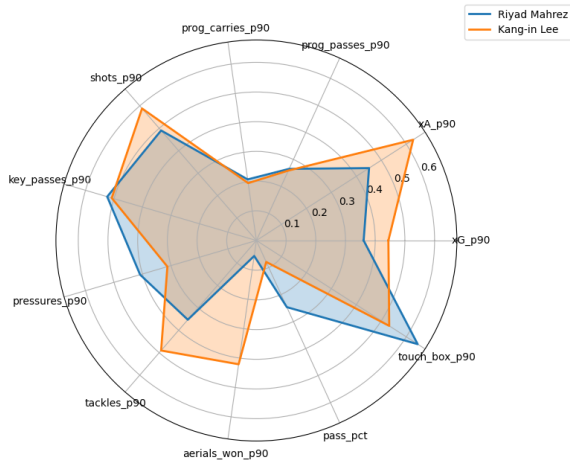
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Mahrez vs Luiz Henrique



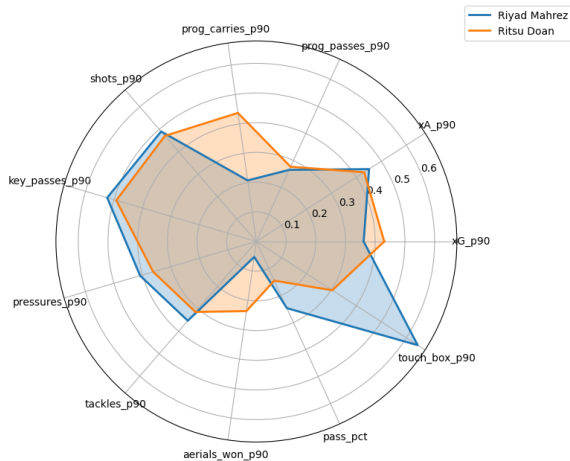
Strong continuity in profile; slight decrease in box touches.

Mahrez vs Kang-in Lee



Higher creation (xA); more central/playmaking tendencies.

Mahrez vs Ritsu Doan

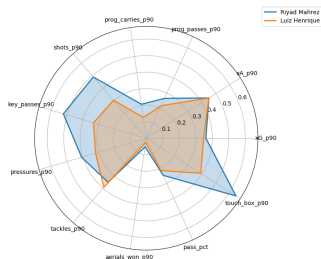


More carries/defensive activity; lower impact in the final third.

Player Profile — Luiz Henrique



Luiz Henrique



Radar vs Mahrez

Basic info

Age : 24 — Left-footed right winger.

Current club : Zenit Saint Petersburg.

Contract : June 2028.

Market value : €25M.

Strengths

Strong continuity with Mahrez's structural profile (0.61).

High xA and key pass volume in right half-space.

Good balance between carries and progressive passes.

Risks

Physical adaptation to league intensity.

Slightly lower box touch volume vs Mahrez.

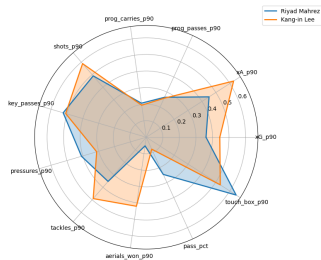
Scout Insight

Similarity profile makes him a plug-and-play option for the Mahrez role.

Player Profile — Kang-in Lee



Kang-in Lee



Radar vs Mahrez

Basic info

Age : 24 — Left-footed wide playmaker.

Position : Creative midfielder starting from the right.

Strengths

Higher xA and creation volume vs Mahrez baseline.

Elite ball retention under pressure.

Excellent vision and space manipulation.

Risks

Lower presence inside the penalty area.

More central positioning than wide Mahrez role.

Scout Insight

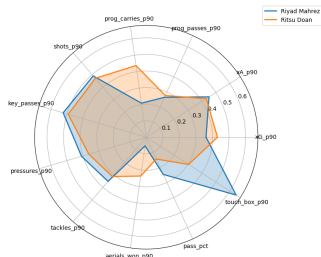
Higher creative upside, but needs tactical adaptation to wide inverted role.

Can raise team playmaking tempo in right half-space.

Player Profile — Ritsu Doan



Ritsu Doan



Basic info

Age : 27 — versatile wide player, capable on both flanks.

High-intensity profile with strong work-rate.

Strengths

Strong ball-carrying and ability to drive forward under pressure.

Significant defensive activity and pressing contribution.

Tactical versatility across wide and inside roles.

Risks / Points of attention

Lower decisiveness in the final third (goals/assists) vs Mahrez.

Less elite-level creation compared to pure playmakers.

Shortlist & Qualitative Analysis

Player	Age	Style	Strengths	Risks
Luiz Henrique	24	Creative, impactful winger	Role continuity, age/upside, tactically transferable	League adaptation
Kang-in Lee	24	Wide playmaker	High xA, excellent game-reading	Less box presence/verticality
Ritsu Doan	27	Versatile wide player	Carries, activity	Lower decisiveness in final third

Recommended Target : Luiz Henrique

Strong multidimensional similarity.

Immediate tactical fit for creation & ball-carrying impact.

Age & upside aligned with club's strategic model.

Player Sheet — Luiz Henrique



Age : 24

Height / Weight : 1.82m — 77 kg

Preferred foot : Left

Primary position : Right winger / wide attacking midfielder

Current club : Zenit Saint Petersburg (Russia)

Contract : June 2028

Market value (Transfermarkt) : €25M

2024/25 production : 14 G/A (9 goals, 5 assists)

Minutes played : 3,490 minutes

Statistical modelling identified profiles **compatible** with Mahrez's role.

Luiz Henrique offers the **best balance** of style, potential, and cost.

Methodology — Dataset & Eligibility Criteria

Position : right winger / wide attacking midfielder

Preferred foot : **natural left-footer** (inverted winger).

Target age range : **18–32 years old**.

Minimum annual playing time : **900 minutes** ; otherwise, the last full season is considered.

Market value (max €25M) : selection limited to players with a credible sporting profile and realistic financial feasibility.

Sample size : roughly 40 players.

Methodology — Selected Indicators (per 90 minutes)

Offensive Impact :

xG_p90 (expected goals)

shots_p90 (shot attempts)

touches_box_p90 (touches inside the penalty area)

Chance Creation / Passing Impact :

xA_p90 (expected assists)

key_passes_p90 (key passes)

pass_accuracy (passing accuracy)

Ball Progression :

progressive_passes_p90 (progressive passes)

progressive_carries_p90 (progressive carries)

Defensive Activity :

pressures_p90 (pressures applied)

tackles_p90 (tackles)

Aerial Game / Duels :

aerials_won_p90 (aerial duels won)

Methodology — Normalization & Comparability

All metrics are averaged over a **90-minute observation window**.

Variable normalization to enable **multidimensional comparison**.

Same advanced data provider (**FBref/StatsBomb**) to ensure consistency and comparability.

Acquisition Cost Proxy

The **Transfermarkt market value** is used as a *proxy* to estimate potential acquisition cost and inform the recruitment strategy.

Methodology — Limitations & Exclusions

Certain leagues (e.g., **Saudi Pro League**, **Süper Lig**, **Ligue 2**, **Denmark**, **Greece**) show **gaps in advanced data coverage**.

These profiles were therefore **excluded** to preserve dataset consistency.

For **Mahrez**, we use **Premier League 2022/23**, his last full season of reference.

Construction of the *Proprietary* Dataset

Manual collection and structuring in Excel, then CSV export.

Merge of FBref (advanced metrics) + Transfermarkt (market value).

Unique, normalized dataset dedicated to this study (fully reproducible via a Python pipeline).

player	team	league	age	position	foot	price_m€	minutes	xG_p90	xA_p90	prog_passes_p90	prog_carries_p90	shots_p90	key_passes_p90	pressures_p90	tackles_p90	aerials_won_p90	pass_pct	touch_box_p90
Riyad Mahrez	Al-Ahli SFC	KSA	34	RW	L	8	1920	0.26	0.23	3.67	3.89	0.55	0.33	11.4	0.23	0.05	88.8	0.68
Lutz Hentges	Zenit St. Petersburg	RUS	24	RW	L	25	2033	0.25	0.23	3.02	2.83	0.36	0.21	9.29	0.25	0.04	79.0	0.39
Kang-In Lee	Paris Saint-Germain	FRA	24	RW	L	25	1663	0.32	0.32	3.5	3.5	0.85	0.32	9.1	0.3	0.19	50.0	0.5
Eden Zhegrea	Juventus Turin	ITA	26	RW	L	25	2288	0.24	0.24	6.2	6.2	0.47	0.24	12.4	0.24	0.24	90.0	0.49
Riccardo Orsolin	FC Bologna	ITA	26	RW	L	25	1876	0.72	0.19	10.1	7.8	0.31	0.58	11.5	0.49	0.16	69.0	0.66
Ritsu Doan	Eintracht Frankfurt	GER	27	RW	L	25	2962	0.31	0.22	3.7	6.8	0.63	0.31	16.3	0.21	0.12	64.0	0.32
Leroy Sané	Galatasaray	TUR	29	RW	L	25	1637	0.6	0.27	6.5	10.2	0.88	0.6	15.8	0.66	0.3	64.0	0.67
David Neres	SSC Napoli	ITA	28	RW	L	24	1263	0.14	0.28	3.5	2.7	0.42	0.14	6.2	0.19	0.25	73.0	0.43
Roger Fernandes	Al Hilal	KSA	19	RW	L	22	1738	0.16	0.16	3.6	3.3	0.31	0.16	6.8	0.17	0.19	56.0	0.36
Leon Bailey	AS Roma	ITA	28	RW	L	22	2956	0.44	0.39	9.2	6.3	0.83	0.44	16.5	0.27	0.4	82.0	0.67
Abu Hadj Moussa	Payenne	NED	23	RW	L	20	1960	0.37	0.14	5.0	2.2	0.5	0.37	8.2	0.15	0.23	96.0	0.37
Quinten Isaksson	Lazio	ITA	24	RW	L	20	2210	0.16	0.06	2.1	6.2	0.24	0.16	7.3	0.21	0.09	46.0	0.3
Doré Luksembi	SL Benfica	POR	28	RW	L	20	3092	0.32	0.06	4.5	16.1	0.38	0.29	13.8	0.29	0.13	93.0	0.42
Rapay	Vasco da Gama	BRA	19	RW	L	18	2301	0.43	0.04	1.6	7.5	0.47	0.35	7.6	0.29	0.06	57.0	0.36
Isaiahoku Fateneu	Leicester City	ENG	21	RW	L	18	1123	0.34	0.16	1.8	1.7	0.4	0.34	3.5	0.14	0.14	41.0	0.28
Ludovic Blas	Stade Rennais	FRA	27	RW	L	18	2293	0.24	0.31	4.6	5.9	0.55	0.24	8.6	0.2	0.18	118.0	0.38
Harry Wilson	Fulham	ENG	28	RW	L	18	1162	0.46	0.08	1.1	4.2	0.54	0.46	5.3	0.33	0.09	51.0	0.41
Emnest Naamati	Lyon	FRA	22	RW	L	15	1092	0.26	0.09	1.5	2.2	0.34	0.26	3.7	0.19	0.13	33.0	0.32
Yasser Asprilla	Girona	ESP	21	RW	L	15	1405	0.19	0.06	2.8	2.1	0.26	0.19	4.9	0.13	0.16	77.0	0.31
Ilan Akhounach	Villarreal	ESP	21	RW	L	15	1511	0.12	0.12	2.6	3.2	0.24	0.12	6.7	0.19	0.15	45.0	0.34
Mohamed-Ali Cho	OGC Nice	FRA	21	RW	L	15	1621	0.17	0.22	4.0	4.0	0.39	0.17	7.9	0.22	0.22	66.0	0.44
Facundo Torres	Palmeiras	BRA	25	RW	L	15	1425	0.25	0.19	2.9	2.9	0.43	0.25	5.8	0.18	0.18	78.0	0.36
Viktor Tsygankov	Girona FC	ESP	27	RW	L	15	1913	0.09	0.24	3.9	2.8	0.33	0.09	6.7	0.13	0.18	65.0	0.31
Andreas Skov Olsen	VfL Wolfsburg	GER	25	RW	L	14	1136	0.63	0.32	3.1	5.3	0.95	0.63	8.4	0.42	0.25	58.0	0.67
Romain Essa	Crystal Palace	ENG	20	RW	L	12	2018	0.18	0.04	4.4	2.2	0.22	0.18	6.6	0.1	0.2	73.0	0.29
Cyprien Ishi	VfB Stuttgart	GER	20	RW	L	12	1099	0.26	0.26	2.4	1.9	0.49	0.18	4.3	0.22	0.19	44.0	0.36
Anders Dreyer	San Diego FC	USA	27	RW	L	12	3019	0.57	0.51	11.8	13.8	1.07	0.48	25.8	0.51	0.35	142.0	0.86
David Brooks	AFC Bournemouth	ENG	28	RW	L	12	980	0.19	0.0	1.9	2.5	0.19	0.19	4.4	0.24	0.16	35.0	0.41
Jerme Steuckens	KRC Genk	BEL	23	RW	L	11	2538	0.28	0.21	11.5	3.4	0.5	0.11	14.9	0.29	0.41	247.0	0.63
Ben Ketbi	Paris FC	FRA	27	RW	L	10	956	0.47	0.38	1.8	3.1	0.85	0.38	4.2	0.29	0.17	73.0	0.47
Morgan Whittaker	Middlesbrough FC	ENG	24	RW	L	10	960	0.0	0.09	1.9	1.7	0.59	0.0	3.6	0.15	0.18	35.0	0.33
Samuel Chukwueze	Fulham	ENG	26	RW	L	10	925	0.29	0.19	3.0	2.7	0.49	0.29	5.7	0.26	0.29	36.0	0.56
Dennis Man	PSV Eindhoven	NED	27	RW	L	10	1995	0.18	0.18	4.2	5.5	0.35	0.18	9.7	0.25	0.19	60.0	0.44